```
> restart:

  # Compute p(0),...,p(Nmax) by Euler's pentagonal recurrence.
  BuildPartitionTable := proc(Nmax::nonnegint)
      local Ptab, nn, r, g, sign, total;

      Ptab := Array(0..Nmax, datatype=anything, fill=0);
      Ptab[0] := 1;

      for nn from 1 to Nmax do
          total := 0;
          r := 1;
          sign := 1;

          while true do
              g := iquo(r*(3*r-1), 2);

              if g > nn then
                  break;
              end if;

              total := total + sign*Ptab[nn-g];

              if g+r <= nn then
                  total := total + sign*Ptab[nn-g-r];
              end if;

              r := r+1;
              sign := -sign;
          end do;

          Ptab[nn] := total;
      end do;

      return Ptab;
  end proc:

  # Compute N(u,nn) using (2.4) with iota=2.
  # Negative partition arguments contribute zero.
  # This formula also handles u=0 when nn>=1.
  RankCoeff := proc(u::nonnegint, nn::posint, Ptab)
      local r, x, term, total, sign;

      total := 0;
      r := 1;
      sign := 1;
```

```
    while true do
        x := nn-r*u-iquo(r*(3*r-1), 2);

        if x < 0 then
            break;
        end if;

        term := Ptab[x];

        if x >= r then
            term := term-Ptab[x-r];
        end if;

        total := total + sign*term;
        r := r+1;
        sign := -sign;
    end do;

    return total;
end proc:

# Exact number of required determinants with 73 <= n <= H.
Count33Through := proc(H::nonnegint)
    local q, rem3;

    if H < 73 then
        return 0;
    elif H <= 106 then
        return iquo((H-73)*(H-72), 2);
    end if;

    q := iquo(H-7, 3);
    rem3 := irem(H-7, 3);

    return iquo(3*q*(q-1), 2)+(rem3+1)*q-1056;
end proc:

# Verify every required determinant in the chosen n-block.
Verify33 := proc(nFirst::posint, nLast::posint)
    local Ptab, nn, m, bound, left, mid, right,
          det, checked, expected;

    if nFirst < 73 or nLast < nFirst or nLast >= 10^6 then
        error "require 73 <= nFirst <= nLast < 10^6";
    end if;
```

```
    expected := Count33Through(nLast)-Count33Through(nFirst-1);

    printf("Computing p(0),...,p(%a).\n", nLast);
    Ptab := BuildPartitionTable(nLast);
    printf("Partition table ready; %a determinants to check.\n",
           expected);

    checked := 0;

    for nn from nFirst to nLast do
        bound := min(nn-73, iquo(nn-7, 3));

        # An empty m-range needs no comparison.
        if bound >= 1 then
            left := RankCoeff(0, nn, Ptab);
            mid := RankCoeff(1, nn, Ptab);

            for m from 1 to bound do
                right := RankCoeff(m+1, nn, Ptab);
                det := mid^2-left*right;

                if det < 0 then
                    error "FAIL: n=%1, m=%2, determinant=%3",
                          nn, m, det;
                end if;

                checked := checked+1;

                # Reuse the adjacent coefficients.
                left := mid;
                mid := right;
            end do;
        end if;

        if irem(nn, 1000) = 0 or nn = nLast then
            printf("Completed n=%a; checked=%a.\n", nn,
checked);
        end if;
    end do;

    if checked <> expected then
        error "incomplete verification: checked=%1,
expected=%2",
              checked, expected;
    end if;
```

```
      printf("PASS: %a <= n <= %a; checked %a determinants.\n",
             nFirst, nLast, checked);

      return [nFirst, nLast, checked, true];
  end proc:

  # Full range in (3.3).
  # Change only these two bounds to verify a smaller block.
  nFirst := 73:
  nLast := 50000:

  result := Verify33(nFirst, nLast):
Computing p(0),...,p(50000).
Partition table ready; 416540620 determinants to check.
Completed n=1000; checked=163120.
Completed n=2000; checked=660620.
Completed n=3000; checked=1491453.
Completed n=4000; checked=2655620.
Completed n=5000; checked=4153120.
Completed n=6000; checked=5983953.
Completed n=7000; checked=8148120.
Completed n=8000; checked=10645620.
Completed n=9000; checked=13476453.
Completed n=10000; checked=16640620.
Completed n=11000; checked=20138120.
Completed n=12000; checked=23968953.
Completed n=13000; checked=28133120.
Completed n=14000; checked=32630620.
Completed n=15000; checked=37461453.
Completed n=16000; checked=42625620.
Completed n=17000; checked=48123120.
Completed n=18000; checked=53953953.
Completed n=19000; checked=60118120.
Completed n=20000; checked=66615620.
Completed n=21000; checked=73446453.
Completed n=22000; checked=80610620.
Completed n=23000; checked=88108120.
Completed n=24000; checked=95938953.
Completed n=25000; checked=104103120.
Completed n=26000; checked=112600620.
Completed n=27000; checked=121431453.
Completed n=28000; checked=130595620.
Completed n=29000; checked=140093120.
Completed n=30000; checked=149923953.
Completed n=31000; checked=160088120.
Completed n=32000; checked=170585620.
Completed n=33000; checked=181416453.
Completed n=34000; checked=192580620.
Completed n=35000; checked=204078120.
Completed n=36000; checked=215908953.
Completed n=37000; checked=228073120.
Completed n=38000; checked=240570620.
Completed n=39000; checked=253401453.
Completed n=40000; checked=266565620.
Completed n=41000; checked=280063120.
Completed n=42000; checked=293893953.
Completed n=43000; checked=308058120.
Completed n=44000; checked=322555620.
```

```
Completed n=45000; checked=337386453.
Completed n=46000; checked=352550620.
Completed n=47000; checked=368048120.
Completed n=48000; checked=383878953.
Completed n=49000; checked=400043120.
Completed n=50000; checked=416540620.
PASS: 73 <= n <= 50000; checked 416540620 determinants.
>
```